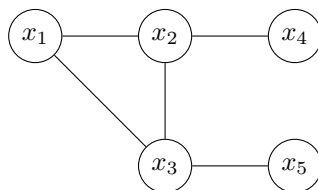


## CS 791: Practice questions on variation optimization and VAE

1. Assume there are two coins with bias  $\theta_1, \theta_2$  respectively. In each trial, a user picks one of the two coins with equal probability, and reports the outcome of tossing the selected coin, which could be H or T. This experiment is repeated 100 times and 60% heads are found. The coin that is selected in each toss is hidden from view, only the outcome (head or tail) is observed. Cast the task of estimating  $\theta_1, \theta_2$  as a variational optimization problem. Exploit symmetry whenever possible to reduce the number of auxiliary variables in a variational form.
  
2. Let  $\mathbf{x} = [x_1, x_2, x_3]$  and  $z$  be real random variables where  $p(z) \sim N(0, 1)$ . Also, we know that  $[x_1, x_2, x_3] = [3z + 2, 4z + 1, z - 2] + \alpha[\epsilon_1, \epsilon_2, \epsilon_3]$  where  $\epsilon_i \sim N(0, 1)$ . With this knowledge, we will plug in the optimal parameters of a VAE defined over them where encoder  $q_\phi(z|\mathbf{x}) \sim N(\mu_{z|\mathbf{x}}, \sigma_{z|\mathbf{x}})$  and decoder  $p_\theta(\mathbf{x}|z) \sim N(\mu_{\mathbf{x}|z}, \Sigma_{\mathbf{x}|z})$ . Assume both encoder and decoder are linear layers where  $\mu_{\mathbf{x}|z} = [\theta_1 z + \theta_2, \theta_3 z + \theta_4, \theta_5 z + \theta_6]$  and  $\mu_{z|\mathbf{x}} = \phi_1 x_1 + \phi_2 x_2 + \phi_3 x_3 + \phi_4$ .
  - (a) Provide the optimal parameters  $\theta_1 \dots \theta_6$  of the decoder mean. ..1 These will match the corresponding coefficients defined in  $[x_1, x_2, x_3] = [3z + 2, 4z + 1, z - 2]$
  - (b) Provide the optimal parameters of the encoder means  $\phi_1 \dots, \phi_4$  ..3 Since the components of  $\mathbf{x}$  are independent of each other given  $z$ , we can estimate  $q(z|\mathbf{x})$  as  $1/3[(x_1 - 2)/3 + (x_2 - 1)/4 + (x_3 + 2)]$  This when simplified will give us values for  $\phi_1, \phi_2, \phi_3, \phi + 4$ .
  - (c) What is the ideal NN to estimate  $\Sigma_{\mathbf{x}|z}$  and  $\sigma_{z|\mathbf{x}}$  from their respective inputs? ..3 The variance  $\sigma_{z|\mathbf{x}}$  to be reliably estimated requires the sample itself and square of individual samples. Thus the  $\sigma$  logic in the encoder must depend on  $x_i$  and square of  $x_i$ . The variance of each  $x_i$  is a constant function of its input and thus any linear layer with a bias parameter could suffice.
  
3. Consider a VAE for which prior  $p(z)$  is Bernoulli distributed with parameter  $\alpha$ . The  $\alpha$  now needs to be learned jointly with the encoder parameter  $\phi$  and decoder parameters  $\theta$ . What will be the function at the last layer of the encoder? Let  $e_i$  denote the output from the last encoder layer. Write the formulae for the  $D_{KL}(q_\phi(z|x^i)||p_\alpha(z))$  in terms of  $\alpha$  and  $e_i$ . ..2 Applying the formula for the KL distance between two Bernoulli distributions, one with parameter  $e_i$  and second with parameter  $\alpha$  gives us the answer as:  $e_i \log \frac{e_i}{\alpha} + (1 - e_i) \log \frac{1 - e_i}{(1 - \alpha)}$
  
4. Consider a social network  $G$  below among five individual where we are interested in inferring the age of the individuals. Let  $x_1, \dots, x_5$  denote the five age variables which we assume are whole integers in the range 20 to 25 (inclusive).



Based on clues present in the profiles of each person, it was possible to define the node potentials for age as follows:

$$\begin{aligned}\psi_1(x_1) &= \exp(-|x_1 - 21|); \quad \psi_2(x_2) = \exp(-|x_2 - 24|); \\ \psi_3(x_3) &= \delta(x_3, 21), \quad \psi_4(x_4) = \delta(x_4, 25), \quad \psi_5(x_5) = \delta(x_5, 22)\end{aligned}$$

where  $\delta(a, b) = 1$  if  $a = b$  and 0 otherwise.

Let the edge potentials be defined as follows:

$$\psi(x_i, x_j) = \begin{cases} \theta_{|x_i - x_j|} & |x_i - x_j| \leq 2 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The joint probability distribution over the five age variables is thus:

$$P(x_1, \dots, x_5) = \frac{1}{Z} \prod_{i=1}^5 \psi_i(x_i) \prod_{(i,j) \in \text{Edge}(G)} \psi(x_i, x_j) \quad (2)$$

Note, this distribution is not necessarily positive since some of the potentials take a value of zero.

We will run the EM algorithm to estimate the edge potential parameters. Assume the observed training data has  $x_1, x_2$  missing and there is only one training instance  $D = \{(x_3 = 21, x_4 = 25, x_5 = 22)\}$ . Assume initial values of the parameters are  $\theta_0^t = 1, \theta_1^t = 2, \theta_2^t = 4$ .

- (a) Write the values of auxiliary  $q$  variables computed in the E-step for the above estimation task. You need to give the exact numerical value of all required  $q$  variables, and not just leave them as formulae. [You only need to show  $q$  variables with non-zero values. Exploit zero potentials to cut down the search space.] ..3 We need to compute

$q_{1z} = P(\mathbf{z} | \theta_0^t = 1, \theta_1^t = 2, \theta_2^t = 4, x_3 = 21, x_4 = 25, x_5 = 22)$  where  $z_1, z_2$  denote the combination of values of  $x_1, x_2$ . First observe that  $P(x_1, x_2 | \theta_0^t = 1, \theta_1^t = 2, \theta_2^t = 4, x_3 = 21, x_4 = 25, x_5 = 22)$  is non-zero only for  $x_2 = 23, x_1 \in \{21, 22, 23\}$ . Thus, we only need to compute these as:

$$P(x_1, x_2 = 23 | \theta_0^t = 1, \theta_1^t = 2, \theta_2^t = 4, x_3 = 21, x_4 = 25, x_5 = 22) \propto \exp(-|x_1 - 21|) \theta_{|x_1 - 21|}^t \theta_{|x_1 - 23|}^t$$

$$q_1(21) \propto \theta_0^t, \theta_2^t = 4 \quad q_1(22) \propto \exp(-1) \theta_1^t, \theta_1^t = 4 \quad q_1(23) \propto \exp(-2) \theta_0^t, \theta_2^t = 4$$

- (b) Write the value of  $Z$  for the above graphical model in terms of  $\theta$  in as simple terms as possible. Again, you will get marks only if you exploit the special structure of the potentials to simplify the computation. ..2

Using the same argument as above, the only possibilities with non-zero scores are  $x_2 = 23, x_1 \in \{21, 22, 23\}, x_3 = 21, x_4 = 25, x_5 = 22$ , thus the value of  $Z$  is just:

$$Z = \sum_{x_1 \in \{21, 22, 23\}} \exp(-|x_1 - 21|) \theta_{|x_1 - 21|} \theta_{|x_1 - 23|} \exp(-1) \theta_2 \theta_1 \theta_2$$

- (c) Give the  $\theta_0$  update equation in the  $M$  step. ..3 In the  $M$  step we need to

$$\max_{\theta} \sum_{x_1 \in \{21, 22, 23\}} q_1(x_i) \log(\exp(-|x_1 - 21|) \theta_{|x_1 - 21|} \theta_{|x_1 - 23|} \exp(-1) \theta_2 \theta_1 \theta_2) - \log Z$$

For computing gradient wrt to  $\theta_0$  the only relevant terms are

$$q(21) \log \theta_0 + q(22) \log \theta_0 - \log Z$$

We compute this using gradient descent.